Foundations of Data Science Final Project

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1 Introduction

Breast cancer is a heterogeneous disease with substantial variation in molecular characteristics and gene expression across tumors. The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) project provides genomic and clinical data that can be used to investigate these molecular differences in breast cancer [1]. In this project, RNA sequencing (RNA-seq) gene expression count data and corresponding clinical and molecular metadata from the TCGA-BRCA cohort were analyzed.

ERBB2 was selected as the primary gene of interest for this analysis. ERBB2 encodes human epidermal growth factor receptor 2 (HER2), a receptor tyrosine kinase with an important role in the development and treatment of breast cancer. Increased HER2 expression is characteristic of a clinically important subset of breast cancers and has led to the development of multiple HER2-targeted therapies [2].

My clinical interest in ERBB2 is also related to my work in pulmonary medicine. HER2-targeted antibody-drug conjugates, including trastuzumab deruxtecan, are associated with interstitial lung disease (ILD) and drug-induced pneumonitis [3]. As a pulmonologist with a clinical focus on ILD, I encounter pulmonary toxicities associated with cancer therapies in clinical practice. This connection between ERBB2-targeted therapy and lung toxicity provided additional motivation for selecting ERBB2 as the primary gene for this analysis.

2 Methods

RNA-seq gene expression count data and corresponding clinical and molecular metadata from the TCGA-BRCA cohort were analyzed [4] using R version 4.6.1. Data manipulation and summary statistics were performed using tidyverse version 2.0.0, including dplyr version 1.2.1. Data visualizations were generated using ggplot2 version 4.0.3. ComplexHeatmap version 2.28.0 and circlize version 0.4.18 were used to generate the gene expression heatmap.

The RNA-seq count matrix contained gene-level expression counts, with genes represented by Ensembl identifiers and samples represented by TCGA

sample identifiers. ERBB2 was selected as the primary gene of interest, and descriptive statistics including the mean, median, standard deviation, minimum, and maximum were calculated for ERBB2 expression. CDK4 was selected as a second gene for comparison with ERBB2. Gene expression data were combined with the corresponding clinical and molecular metadata using TCGA sample identifiers. PAM50 molecular subtype was selected as the primary categorical covariate, and age at initial pathologic diagnosis was selected as an additional clinical covariate.

A histogram was used to examine the distribution of ERBB2 expression, with the mean and median indicated by reference lines. A scatter plot was used to compare ERBB2 and CDK4 expression, with a linear regression line added to visualize their relationship. Because the RNA-seq counts were right-skewed and spanned a wide range of values, log10 scales were used for the gene expression axes. ERBB2 expression across PAM50 molecular subtypes was examined using a violin plot with an overlaid box plot and a cumulative distribution function (CDF). Finally, a heatmap was generated using a panel of 10 breast cancer-related genes. For the heatmap, raw counts were transformed using $\log2(\text{count} + 1)$, and both genes and samples were hierarchically clustered. PAM50 molecular subtype was included as a sample annotation.

3 Results

3.1 Clinical and Molecular Characteristics

Table 1 summarizes the distribution of PAM50 molecular subtypes and age at initial pathologic diagnosis in the TCGA-BRCA cohort. Among samples with available PAM50 classification, Luminal A was the most common subtype (52.0%), followed by Luminal B (19.0%), Basal (17.9%), HER2-enriched (7.5%), and Normal-like (3.6%). The mean age at initial pathologic diagnosis was 58.42 years.

Table 1: Summary of selected clinical and molecular covariates

Covariate	Statistic	Value
PAM50 subtype	Basal, n (%)	197 (17.9%)
	HER2-enriched, n (%)	82 (7.5%)
	Luminal A, n (%)	571 (52.0%)
	Luminal B, n (%)	209 (19.0%)
	Normal-like, n (%)	40 (3.6%)
Age at initial pathologic diagnosis	Mean	58.42 years
	Median	58 years
	Standard deviation	13.15 years
	Minimum	26 years
	Maximum	90 years

3.2 Distribution of ERBB2 Expression

ERBB2 RNA-seq counts demonstrated a right-skewed distribution, with a mean of 46,982.49 counts and a median of 19,125 counts. The standard deviation was 105,287.5 counts, with values ranging from 205 to 995,067 counts. The mean exceeded the median due to a subset of samples with substantially higher expression. A log10 scale was used to better visualize the distribution across the wide range of ERBB2 expression counts (Figure 1).

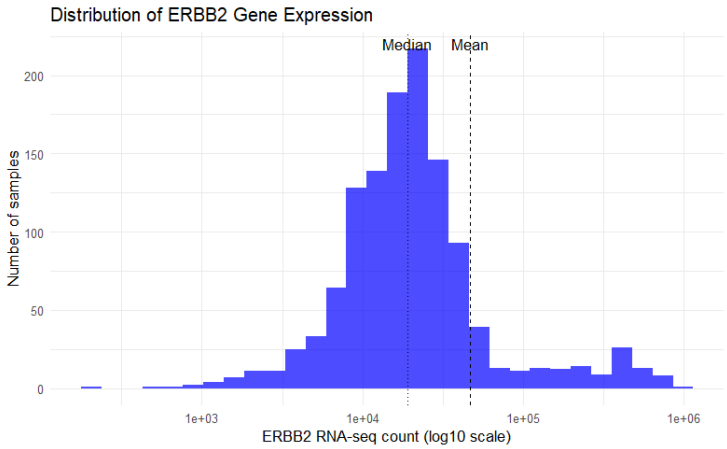


Figure 1: Distribution of ERBB2 RNA-seq counts in the TCGA-BRCA cohort. The x-axis is displayed on a log10 scale. The dashed and dotted vertical lines represent the mean and median, respectively.

3.3 Relationship Between ERBB2 and CDK4 Expression

ERBB2 and CDK4 expression demonstrated a modest positive relationship, although substantial variability was present across samples. Log10 scales were used for both axes because of the wide range of RNA-seq counts. A fitted linear regression line was included to visualize the overall relationship between the two genes (Figure 2).

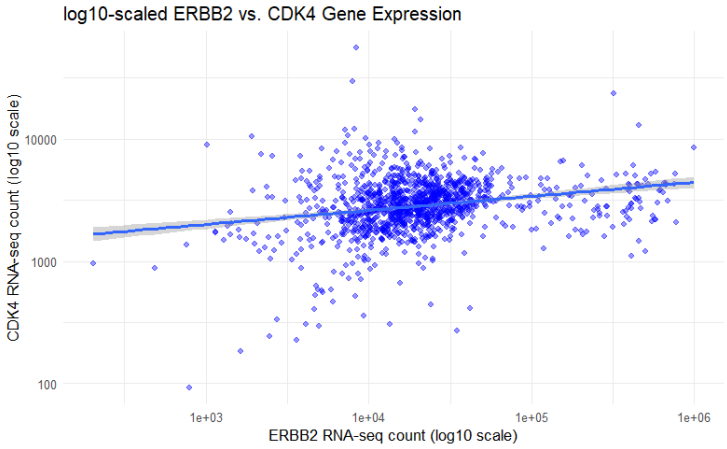


Figure 2: Relationship between ERBB2 and CDK4 RNA-seq counts in the TCGA-BRCA cohort. Both axes are displayed on log10 scales, and the fitted line represents a linear regression trend.

3.4 ERBB2 Expression Across PAM50 Molecular Subtypes

ERBB2 expression differed across PAM50 molecular subtypes. The HER2-enriched subtype demonstrated substantially higher ERBB2 expression overall, although considerable variability was present within the subtypes. The violin plot displays the distribution of ERBB2 expression, while the overlaid box plots show the median and interquartile range (Figure 3).

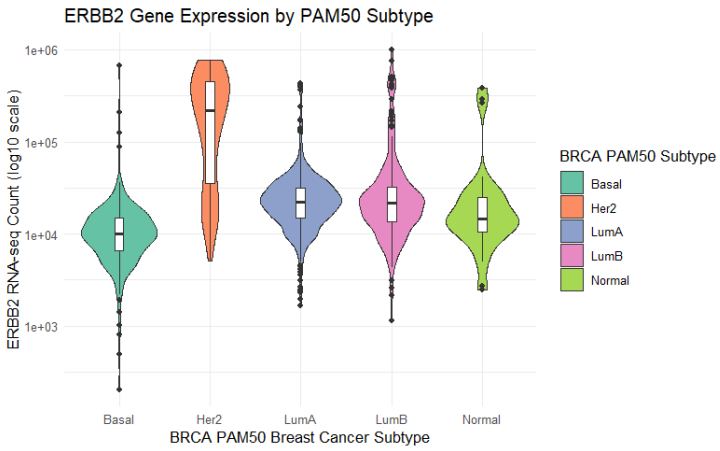


Figure 3: Distribution of ERBB2 RNA-seq counts across PAM50 molecular subtypes in the TCGA-BRCA cohort. ERBB2 expression is displayed on a log10 scale. Violin plots represent the distribution of expression, and the overlaid box plots display the median and interquartile range.

3.5 Cumulative Distribution of ERBB2 Expression

The cumulative distribution of ERBB2 expression further demonstrated differences among PAM50 molecular subtypes. The HER2-enriched subtype showed a rightward shift in the distribution, indicating generally higher ERBB2 expression compared with the other subtypes. Basal tumors generally demonstrated lower ERBB2 expression, while the remaining subtypes showed intermediate distributions (Figure 4).

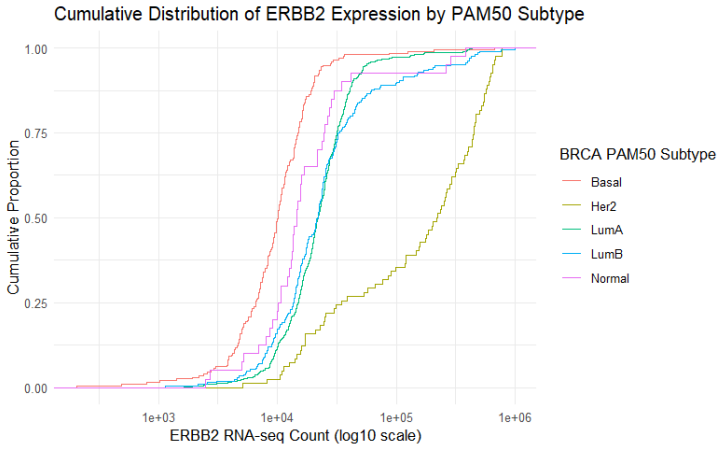


Figure 4: Cumulative distribution of ERBB2 RNA-seq counts across PAM50 molecular subtypes in the TCGA-BRCA cohort. ERBB2 expression is displayed on a log10 scale.

3.6 Gene Expression Heatmap

The heatmap demonstrated distinct patterns of gene expression across the TCGA-BRCA samples and PAM50 molecular subtypes. ERBB2 expression was higher in many samples classified as HER2-enriched. Other subtype-associated patterns were also visible, including lower ESR1 and PGR expression and higher KRT5 and EGFR expression in many Basal samples. Hierarchical clustering grouped samples and genes according to similarities in their expression patterns (Figure 5).

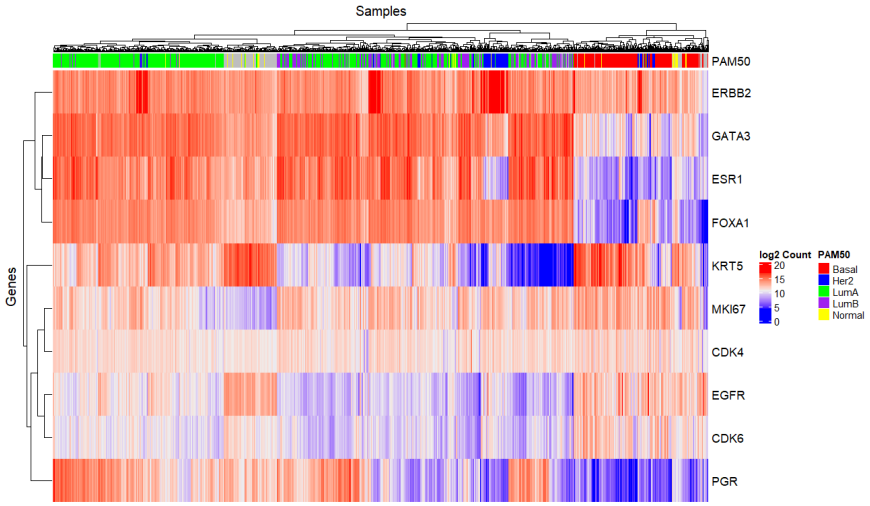


Figure 5: Heatmap of 10 breast cancer-related genes across TCGA-BRCA samples. RNA-seq counts were transformed using log2(count + 1). Rows represent genes and columns represent samples. Hierarchical clustering was performed for both genes and samples, and PAM50 molecular subtype is shown as the sample annotation.

Acknowledgments

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References

[1] The Cancer Genome Atlas Network. (2012). Comprehensive molecular portraits of human breast tumours. *Nature, 490*(7418), 61–70. doi:10.1038/nature11412.

[2] Gutierrez, C., & Schiff, R. (2011). HER2: Biology, detection, and clinical implications. *Archives of Pathology & Laboratory Medicine, 135*(1), 55–62. doi:10.5858/2010-0454-RAR.1.

[3] Tarantino, P., Modi, S., Tolaney, S. M., Cortés, J., Hamilton, E. P., Kim, S.-B., Toi, M., Andrè, F., & Curigliano, G. (2021). Interstitial lung disease induced by anti-ERBB2 antibody-drug conjugates: A review. *JAMA Oncology, 7*(12), 1873–1881. doi:10.1001/jamaoncol.2021.3595.

[4] National Cancer Institute. (2026). TCGA Breast Invasive Carcinoma (TCGA-BRCA). *Genomic Data Commons Data Portal*. Accessed October 2026.